

Julian Voss

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SUMMARY

Embedded Systems Engineer with 6 years developing firmware for real-time and low-power devices. Skilled in C, C++, and RTOS. Cut power consumption by 35% on a flagship IoT product.

TECHNICAL SKILLS

Languages: C, C++, Rust

RTOS: FreeRTOS, Zephyr

Hardware: ARM Cortex-M, STM32, I2C/SPI/UART

Tools: JTAG debugging, oscilloscopes, logic analyzers

Practices: Low-power design, memory-constrained optimization

PROFESSIONAL EXPERIENCE

Senior Embedded Systems Engineer — Alderwood IoT, San Jose, CA

May 2020 – Present

- Developed firmware in C for an ARM Cortex-M4-based sensor product shipping 500K+ units annually
- Redesigned power management routines, cutting average power consumption by 35% and extending battery life to 2 years
- Ported firmware from a bare-metal loop to FreeRTOS, improving task scheduling reliability under high sensor load
- Debugged and resolved a critical field failure using JTAG and logic analyzer traces, preventing a product recall

Firmware Engineer — Cascade Robotics, San Jose, CA

Jun 2018 – May 2020

- Wrote low-level drivers for I2C and SPI peripherals on custom robotics control boards
- Optimized memory usage in a 64KB RAM-constrained system, freeing headroom for new features

PROJECTS

cold_chain_engine — Rust, Cortex-A78 (bare-metal)

- Bare-metal Rust engine for cold-chain sensor monitoring with manual memory management and no_std architecture

EDUCATION

B.S. Electrical Engineering — San Jose State University, 2018

CERTIFICATIONS

- ARM Accredited Engineer — ARM, 2021